

1.

Variable:

$$(i=1, n) \sum 1/i^2 \leq 2 - 1/n \text{ for all } n \geq 1$$

i is the variable that induction is being performed on.

Base Case: $n = 1$

$$(i=1, 1) \sum 1/i^2 \leq 2 - 1/1$$

$\rightarrow 1 \leq 1$ so the base case holds and $P(1)$ is true.

Induction Hypothesis:

Assume that the defined equation stands for an n value where $n = k \geq 1$, where

$$(i=1, k) \sum 1/i^2 \leq 2 - 1/k$$

Use this induction hypothesis to prove the statement is correct and stands true for all applicable values. $N = k+1$, the inductive step, is proven next

Inductive Step:

Prove that $n = k+1$ applies as true.

$$(i=1, k+1) \sum 1/i^2 \leq 2 - 1/(k+1)$$

$$(i=1, k+1) \sum 1/i^2 = (i=1, k) \sum 1/i^2 + 1/(k+1)^2$$

- Split apart the summation with the $k+1$ in it.

$$(i=1, k+1) \sum 1/i^2 \leq 2 - 1/k + 1/(k+1)^2$$

- Plug in the inductive hypothesis

$$2 - 1/k + 1/(k+1)^2 \leq 2 - 1/(k+1)$$

- Set summation equal to inductive hypothesis counterpart in equation.

$$-1/k + 1/(k+1)^2 \leq -1/(k+1)$$

- Cancel out the 2s.

$$1/(k+1)^2 \leq 1/k - 1/(k+1)$$

- Add $1/k$ to each side.

$$1/k - 1/(k+1) = 1/k(k+1)$$

- Rewrite right side.

$$1/(k+1)^2 \leq 1/k(k+1)$$

$$k \leq k+1$$

- Multiply both sides by $k(k+1)^2$ to get this simple statement.

- The above inequalities holds true for all $k \geq 1$, therefore the statement

$(i=1, k+1) \sum 1/i^2 \leq 2 - 1/(k+1)$ is proven true and the inductive step is proven true.

- $n = k+1$, and plugging in n for k will not change anything regarding its nature.

By induction, $(i=1, n) \sum 1/i^2 \leq 2 - 1/n$ is true for all $n \geq 1$.

$$\forall n \in \mathbb{N} \text{ and } n \geq 1, (i=1, n) \sum 1/i^2 \leq 2 - 1/n$$

2.

Show that $\mathbb{N}$ and $\mathbb{R}$ are equinumerous (bijection)

$$\mathbb{Q} = \{p/q \mid p \in \mathbb{Z}, q \in \mathbb{N}, \gcd(p,q) = 1\}$$

- All rational numbers can be expressed as fractions ($\mathbb{R}$)

$\mathbb{Q}^+ = \{p/q \mid p, q \in \mathbb{N}\}$ and iterate through the set with p and $q \neq 1$ where all of the p and q combos are in a diagonal pattern

- For example, the first diagonal in this imaginary grid is just $1/1$ where p and $q = 1$. Then in next diagonal, it would be $2/1$ and $1/2$. Then $3/1, 2/2, 1/3$.
- These diagonals ensure that there are no repeating numbers, as p reduces from the n maximum number while q increases from 1 in the n th diagonal.

Now include the $\mathbb{Q}^-$ elements regarding the diagonals, which still ensures no repeating elements.

- Ex: $0, 1/1, -1/1, 1/2, -1/2, 2/1, -2/1, \dots \in \mathbb{Q}$

Bijection function $f: \mathbb{N} \rightarrow \mathbb{Q}$ with the first element $f(1) = 0$, and $f(n, n+1, n+2, \dots)$ being the n elements where $n \in \mathbb{Q}$ excluding 0.

- So $f(1) = 1/1, f(2) = -1/1$, and so on.

This is a bijection as every rational number appears once and every natural number has a unique rational number. Therefore $\mathbb{N} \sim \mathbb{Q}$, $\mathbb{N}$ and $\mathbb{Q}$ are equinumerous.

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

$$\mathbb{Q} = \{p/q \in \mathbb{R} \mid p \in \mathbb{Z}, q \in \mathbb{N}\} \text{ and } p/q \text{ is in the lowest numerator and denominator.}$$

Prove $\exists f: \mathbb{N} \rightarrow \mathbb{Q}$ where f is bijection

$$\mathbb{Q}^+ = \{p/q \in \mathbb{Q} \mid p, q \in \mathbb{N}\} \text{ and } p/q \text{ is in divided form}$$

Let $M = \{r_1, r_2, r_3, r_4, \dots\} \subseteq \mathbb{Q}^+$, where $r_i = p_i/q_i$ is in lowest divided form and occurs only once.

Can iterate diagonally through the $\mathbb{N} \times \mathbb{N}$ grid where $(p, q) \in \mathbb{N} \times \mathbb{N}$ and ignore the non-reduced, non lowest form fractions so that $\exists f^+: \mathbb{N} \rightarrow \mathbb{Q}^+$ so f^+ is bijective

$$\mathbb{Q} = \mathbb{Q}^+ \cup \{0\} \cup \{-r \mid r \in \mathbb{Q}^+\}$$

Insert 0 at the beginning and combine the $\mathbb{Q}^+$ and $\mathbb{Q}^-$ versions of the diagonal values

Injectiveness: Assume $f(n_1) = f(n_2)$, then n_1 and n_2 map to same rational in form $r_{k1}, -r_{k1}$

Since all r_k are unique due to lowest terms and diagonalization, along with binary sign, then $n_1 = n_2$ is implied.

Surjectiveness: $q \in \mathbb{Q}$, then one of conditions must be upheld:

$$q = 0, \text{ and } f(1) = 0$$

$$q > 0, \text{ and } \exists k \in \mathbb{N} \text{ where } r_k = q \rightarrow f(2k) = q$$

$$Q < 0, \text{ and } \exists k \in \mathbb{N} \text{ where } r_k = -q \rightarrow f(2k+1) = q$$

Every $q \in \mathbb{Q}$ is in range of f , therefore f is surjective.

$\exists f: \mathbb{N} \rightarrow \mathbb{Q}$ such that f is bijection $\rightarrow \mathbb{N} \sim \mathbb{Q}$. Therefore, $\mathbb{N}$ and $\mathbb{Q}$ are equinumerous

3. Prove that if S_1 is countable and S_2 is uncountable, then $S_2 \setminus S_1$ is uncountable.

Let S_1 be countable set

- $\exists f_1: \mathbb{N} \rightarrow S_1$ such that f_1 is a surjection

S_2 is uncountable set.

- $\exists f_2: \mathbb{N} \rightarrow S_2$ so f_2 is surjection

Contradictively assume that $S_2 \setminus S_1$ is countable

- $\exists f_3: \mathbb{N} \rightarrow (S_2 \setminus S_1)$ so f_3 is a surjection

Combine S_1 and $S_2 \setminus S_1$ together into one set

$$S_2 = (S_2 \cap S_1) \cup (S_2 \setminus S_1)$$

$S_1 \subseteq S_1$ and $S_2 \cap S_1 \subseteq S_1$, therefore S_1 is countable

$S_2 \setminus S_1$ is countable through the contradiction assumption.

And as shown above, $S_2 \cap S_1 \subseteq S_1$ is countable while $S_2 \setminus S_1$ is contradictively assumed to be countable.

$$S_2 = (S_2 \cap S_1) \cup (S_2 \setminus S_1)$$

- Both sides of this union then are countable, and if two sets are countable, then the union between the two sets are also countable
- THEREFORE, S_2 is countable!!!
- **CONTRADICTION!** This contradicts the given statement in the problem that S_2 is uncountable.
- Assumed that S_1 is countable, S_2 is uncountable, and $S_2 \setminus S_1$ is countable. This then guided to show that S_2 is countable, which contradicts the problem statement.

Through contradiction, the assumption that $S_2 \setminus S_1$ is countable is false. Therefore $S_2 \setminus S_1$ is uncountable in this situation.

$\forall S_1, S_2$ where S_1 is countable $\wedge S_2$ is uncountable $\rightarrow S_2 \setminus S_1$ is uncountable

4A. Given regular expression, simplified as much as possible, for language of all strings of a's, b's, and c's where a is never followed by another b.

Conditions: a can be followed by another a or c, b and c can be followed by any character, a can be the last character.

- So all strings in the language where substring 'ab' does not occur.

$$\Sigma = \{a, b, c\}$$

$\forall$ instance of 'a' in string, $\neg$ followed by 'b'

$$L = \{w \in \{a, b, c\}^* \mid \forall j \text{ if } j \text{ is not last character, if } w_j = a \text{ then } w_{j+1} \neq b\}$$

- Standalone b or c is always allowed, a can be at end of string or only followed by a or c.

$$A = (a(a|c))$$

$$B = b$$

$$C = c$$

$$((b|c) \mid a(a|c))^*(\epsilon \mid a)$$

$$L = L(((b|c) \mid a(a|c))^*(\epsilon \mid a))$$

Creates all strings in $L = \{a,b,c\}$ so no 'a' is followed by a 'b'

(b|c): Any b or c has no ruling.

a(a|c): a can be followed by another a or c

*: Can be repeated for any combination of string.

($\epsilon \mid a$): String can end with either nothing or another a.

4B. Simplify $(a+b)^*(a+b+c)^*(a+b)^*$ and describe concisely in English the language described.

$$R = (a+b)^*(a+b+c)^*(a+b)^*$$

$(a+b)^* \rightarrow$ all strings constructed from 'a' and 'b'

$(a+b+c)^* \rightarrow$ all strings made from 'a', 'b', and 'c'

Both conditions also include ϵ

$$L1 = (a+b)^*$$

$$L2 = (a+b+c)^*$$

This splitting of the language then turns the expression into $L1 \cdot L2 \cdot L1$

But $L1 \subseteq L2$, so then can be simplified to:

$$L1 \cdot L2 \cdot L1 \subseteq L2 \cdot L2 = L2$$

It can be simplified to $L2$, as any string in $L2$ can be written as some a's and b's and then any combination of a's, b's, and c's. Since $L1$ comprises of some a's and b's, it is a subset of $L2$ and can be simply written as $L2$.

$$(a+b+c)^*$$

Language consists of all strings made up of any amount of a's, b's, and c's in any order, whilst including the empty string. All strings over the alphabet $\{a, b, c\}$

4C. Simplify $(a^*abb) + (ba^*bb)$ with smallest amount of operators.

This is the union of two strings that both end with 'bb' but the (ba^*bb) string has a 'b' followed by any amount of a's first while the (a^*abb) has just any amount of a's and a single 'a' before the ending.

- So both end in 'bb' and have any amount of a's as well.

Rewrite to $x + a^* + bb$ where $x \in \{\epsilon, b\}$

$(a^*abb) + (ba^*bb) = (a^*abb) + (ba^*bb)$ (Rewriting for clarity)

- Everything contains a^*bb , its a common factor. Amount of a's followed by two b's.
- Factor the two b's

$a^*abb = (a^*a)bb = a^+bb$

- $a^+ = a^*a$ | at least one a is guaranteed.

$a^+bb + ba^*bb = (a^+ + ba^*)bb$

- Factor out the two b's (bb) and this is the final expression.

$(a^+ + ba^*)bb$

In alphabet $\{a, b\}$, it is every string that ends in 'bb' and either has one or more a's or one b followed by any number of a's beforehand.

$L = \{x \in \{a, b\}^* \mid x \text{ needs to end in } bb \text{ and starts with either } a^+ \text{ or } ba^*\}$

$L = L((a^+ + ba^*)bb)$

$\subseteq \wedge \in \forall \exists \neg \Sigma \epsilon \cup$